



NEW PRODUCTS

MAKE IN INDIA

TEST & MEASUREMENT

Thermal imaging camera

FLIR Systems, Inc. has introduced the ETS320 thermal imaging solution for electronics testing in engineering benchtop environments. This



product combines a high-sensitivity thermal camera with an adjustable, hands-free table stand to provide consistent, non-contact thermal testing. By offering more than 76,000 points of temperature measurement, it offers the ability to monitor power consumption, detect hot spots, and identify potential points of failure during product development.

This rechargeable, battery-powered unit can be connected to a PC running FLIR Tools software for detailed data analysis, recording and reporting.

FLIR Systems, Inc.
www.flir.in/ETS320

Gas sensor

Toshniwal Sensing Devices has introduced CO₂ gas sensor module CDM7160 from FIGARO, Japan. This sensor can be used in



indoor air-quality control systems, including variable-air volume systems and demand-controlled ventilation systems. It uses a compact NDIR CO₂ sensor, providing high accuracy at low power. Two detector elements inside the module make absolute measurement possible.

The module comes with both a UART and I2C digital interface. Evaluation module EM7160 is also available for easy testing of the CDM7160. It provides CO₂ measurement data output in serial data format via a USB connector, pulse-width modulated (PWM) output and continuous volt-

age output (converted from the PWM output signal).

Toshniwal Sensing Devices Pvt Ltd
www.tsdp.com

WLAN test software

National Instruments (NI) has introduced WLAN Test Toolkit 17.0 with support for Draft 1.1 of the IEEE



802.11ax standard. Combined with NI's second-generation vector signal transceiver, the toolkit supports 802.11ax waveform generation and analysis for characterisation, validation and production test of products such as RF front-end components, wireless modules and user devices.

The toolkit helps users solve new access point test cases by generating signals that simulate multiuser environments, including per-user impairments.

National Instruments
www.ni.com/80211ax/

COMPONENTS

RF synthesisers

Analog Devices' 13.6GHz ADF5356 wide-band synthesiser with integrated voltage-controlled oscillator (VCO) is suited for applications like wireless



infrastructure, microwave point-to-point links, electronic test and measurement, and satellite terminals. The ADF4356 is a complementary synthesiser product that operates to 6.8GHz and is comparable in performance.

The new synthesisers generate RF outputs from 53.125MHz to 13.6GHz without gaps in frequency coverage.

These feature VCO phase noise of -113dBc/Hz at 100kHz offset at 5GHz, with integrated rms jitter of just 97 fsec (1kHz to 20MHz).

Analog Devices, Inc.
www.analog.com

Silicon-carbide diodes

STMicroelectronics has introduced a full range of 2A-40A 1200V silicon-carbide (SiC) JBS (Junction Barrier Schottky) diodes. Using SiC technology, the new diodes are said to offer high switching efficiency, fast recovery and consistent temperature characteristics.



Featuring a lower forward voltage, these are suited for cost-conscious applications including solar inverters, industrial motor drives, home appliances and power adaptors.

At the same time, performance-oriented applications can use these for superior efficiency, low weight, small size or best thermal properties. The higher efficiency provided by lower forward voltage drop delivers important benefits for automotive equipment such as on-board battery chargers and charging stations for plug-in hybrid or electric vehicles as well.

STMicroelectronics
www.st.com

Radar sensor chips

Texas Instruments (TI) claims to have unveiled the smallest millimetre-wave radar sensor chips that can measure distances smaller than the width of human hair. Utilising CMOS process, the new chips integrate both analogue and digital functions. These find applications in autonomous driving, industrial robotics, medical devices, and factory and building automation.

The sensors include five solutions across two families of 76 to 81GHz